

A Science-Based Ion Transport Simulation Game

CytoTransport

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1 Overview

This game simulates the dynamics of ion transport across a biological cell membrane. It illustrates how physical barriers and protein pore structures regulate the internal environment of a cell. The cell's environment is split into four distinct components:

Intracellular Region: Represents the protein-rich cytoplasm of the cell's interior. Rendered in gray, initially, and changes to red as the ion concentration increases.

Extracellular Region: Represents the aqueous, water-rich environment surrounding the cell. Rendered in light blue.

Cell Membrane: Represents the hydrophobic (i.e., water repelling) phospholipid bilayer that creates the cellular wall, separating the intra- and extracellular regions. Rendered in orange.

Protein Pores: Represent the tapered protein that spans the cellular wall, allowing ions to flow across. Rendered in white.

2 Core Simulation Modules

2.1 The Hydrophobic Lipid Bilayer

The phospholipid bilayer is **hydrophobic** and forms the cell wall. Due to its hydrophobicity, polar molecules (e.g., H_2O) and charged ions (e.g., Na^+ , K^+ , Cl^-) are unable to simply pass through it. Instead, nature created alternative pathways, including protein ion pores. Due to the water-rich intra- and extracellular environments, ions (plus polar molecules and other biopolymers) can move around.

Game Mechanic: The Lipid Bilayer Barrier

Ions interact with the lipid bilayer through collisions and are randomly scattered back.

Scientific Basis: The ion scattering represents the high energy barrier that an ion encounters when attempting to diffuse through the phospholipid's non-polar fatty acid tails. The change in the ion's trajectory reflects the scattering effect of hitting a dense, hydrophobic and heterogeneous molecular cell barrier, and the multitude of intermolecular forces that occur.

Code:

- **Bounce Logic:** `ion.vy = abs(ion.vy)` ensures a downward trajectory.
- **Randomness:** `ion.vx = random.uniform(-2, 2)` simulates slight scattering.

[Image of the phospholipid bilayer structure]

2.2 Channel Architecture: Tapered Pore Proteins, Steric Exclusion and Gating Thresholds

The ion channels are rendered as **light gray** tapered wedges using `pygame.draw.polygon`.

Channel Architecture: Tapered Protein Wedges

The protein structures forming the pores are rendered as oriented 4-point polygons to reflect biological asymmetry.

- **Vestibule Geometry:** The wedges are anchored at the gap boundaries (`gl`, `gr`). They extend into the membrane with a depth of 15px.
- **Internal Taper:** The top points of the wedges (intracellular side) extend a further 6px into the channel than the bottom points.
- **Result:** This creates a physical bottleneck that narrows toward the cell interior.

Scientific Reality (Extracellular Vestibule): The wider bottom represents the water-filled entrance of a channel. This lowers the electrostatic energy barrier, effectively "harvesting" ions from the extracellular space.

Scientific Reality (Selectivity Filter): The narrowing at the top models the **Selectivity Filter**. In channels like K^+ , this narrow region forces ions to shed their **hydration shells** (water molecules) and interact directly with the protein's carbonyl oxygens. This ensures that only ions of the correct size and charge can successfully translocate.

A key refinement in the simulation is the enforcement of physical occupancy limits within the channel.

Game Mechanic: Size-Dependent Collision

The collision logic now evaluates the `ion.rect` boundaries against the pore's `current_gap`.

- **Partial Blockage:** If any part of the ion's width (r) overlaps with the tapered protein wedges, a collision is registered.
- **Gating Limit:** If `PoreWidth` < `IonSize`, the probability of passage drops to zero.

Scientific Reality: This models **Steric Exclusion**. In biological membranes, the Permeability (P) of a channel is not just a binary “open/closed” state but a function of the pore's radius relative to the ion's **hydrated radius**. If the pore constricts below a certain threshold, the energetic cost of dehydration or physical crowding prevents the ion from translocating.

2.3 Ion Channels and Conformational Change

The pores systematically enlarge and shrink, simulating the movement of protein subunits.

Game Mechanic: Sinusoidal Gating

Pore width is determined by a sine wave:

$$\text{CurrentGap} = \text{BaseGap} + \sin(\text{timer}) \cdot \text{Amplitude}$$

Scientific Reality: This simulates **conformational change** or “gating.” Channels respond to various stimuli to open and close their central pore, regulating the rate of flux.

2.4 Ion Characteristics

2.4.1 Ion Selectivity

The simulation allows for multiple ion species, each defined by a unique **Ionic Radius** and **Hydration Shell** proxy.

Game Mechanic: Variable Hitboxes

The `Ion` class accepts a `size` parameter. Larger ions (e.g., K^+) have a wider `pygame.Rect`, making it physically more difficult to pass through the narrowing wedge of the pore during its “constricted” sine-wave phase.

Scientific Reality: This models **Steric Hindrance**. In a real cell, ion channels are highly selective. A channel might be physically too narrow for a large ion to pass, or it may require the ion to shed its “hydration shell” (water molecules sticking to the charge) to fit through the selectivity filter.

2.4.2 Ion Kinematics and Viscosity

The simulation assigns a unique velocity vector to each ion species, creating a non-uniform flux rate.

Table 1: Possible ions. The size of the ions corresponds to their radii (scale: 7px = 1.0Å)

Ion	Charge	Atomic Radius (Å)	Color	Biological Context
Sodium (Na^+)	+1	1.0	Blue	
Calcium (Ca^{2+})	+2	1.0	Yellow	
Potassium (K^+)	+1	1.4	Purple	
Chloride (Cl^-)	-1	1.8	Green	

Game Mechanic: Velocity-Size Scaling

Ion speed is inversely proportional to ion size:

- **Small Ions** (Na^+): High velocity ($v = -6$), mimicking high kinetic energy.
- **Large Ions** (Cl^-): Low velocity ($v = -2$), mimicking high fluid resistance.

Scientific Reality: This reflects the **Stokes-Einstein Relation**. In a viscous medium like the extracellular fluid, larger particles experience greater hydrodynamic drag. This results in a lower diffusion coefficient, meaning they migrate more slowly toward the membrane than smaller particles.

2.5 Concentration Gradients (Visual Feedback)

The background colors provide immediate visual feedback regarding the compartmentalization of the cell.

Game Mechanic: Environmental Contrast

Extracellular (Static): The bottom half of the screen remains **light blue**, reflecting a stable, water-rich environment. **Intracellular (Dynamic):** The top half starts as a **light gray**, indicating a neutral internal environment. As the `intracellular_count` rises, this gray transitions toward red.

Scientific Reality: The color shift from **light blue** to gray highlights that the intracellular and extracellular environments are not identical. The cytoplasm is a crowded environment filled with organelles, proteins, and cytoskeletal structures, which is represented by the initial gray neutral tone, distinct from the **light blue** "water" outside.

While the background color provides visual feedback of the gradient, the physical manifestation of this gradient is handled by the Pressure Factor (see Section 2.5).

2.5.1 The Pressure Factor (Kinetic Logic)

The simulation tracks a `pressure_multiplier` (Π) which acts as a scalar for the velocity vectors of internal particles.

Game Mechanic: Kinetic Scaling

The multiplier Π is calculated as:

$$\Pi = 1.0 + \left(\frac{n_{intracellular}}{25.0} \right) \quad (1)$$

When an ion $\vec{v}$ collides with a boundary, the new velocity $\vec{v}'$ is determined by:

$$\vec{v}' = -(\vec{v} \cdot \Pi) \quad (2)$$

Scientific Reality: This represents **Chemical Potential** (μ). In high-concentration systems, the number of molecular collisions per second increases, effectively increasing the "force" pushing particles back toward the pores.

Table 2: Scaling Reference Table

Ion Count	Multiplier (Π)	Thermodynamic State
0	1.00×	Ground State (Light Gray)
10	1.40×	Low Concentration (Pale Pink)
25	2.00×	High Flux (Salmon Red)
50	3.00×	Saturation/Critical Pressure (Deep Red)

2.6 Mid-Pore Boundary

The intracellular environment is visually and physically represented as extending into the membrane.

Game Mechanic: Mid-Pore Transition

The intracellular background rectangle is drawn to `MEMBRANE_Y + (THICKNESS / 2)`.

Scientific Reality: This represents the **Aqueous Pore**. The internal fluid of the cell partially penetrates the channel protein, allowing ions to enter the "exit vestibule" before fully leaving the membrane.

3 Future Development & Feature Ideas

- **Ion Selectivity:** Introduce different radii for ions (e.g., Ca^{2+} vs Na^+).
- **Relative Position:** Move the cell wall along the y-axis.
- **Modulators:** Add molecules that changes the pore dynamics (e.g., closed, open).
- **Extracellular Proteins:** Add extracellular proteins that float and block the pores access briefly.
- **Intracellular Proteins:** add intracellular organelles that consume the ions.
- **Multiple Ion Types:** Add the ability to cycle through different ions (and thus, their sizes).

- **Ion nature dependency:** Provide a specific "Score Multiplier" logic in the Python code so that successfully transporting a slow, bulky chloride ion gives the player more points than a fast sodium ion? Maybe also think about the role of the formal charge, or the effects of hydration (e.g., sodium ion has a tight solvation shell, making its effect size and speed slower).
- **Electrochemical Gradients:** Program pores with a static charge to attract or repel specific ion types.
- **Active Transport:** Add a "Pump" mechanic requiring energy (ATP) to move ions against the pressure gradient.